

A novel hybrid model for crude oil price forecasting based on MEEMD and Mix-KELM

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ABSTRACT

It is of vital importance for governments, enterprises, and investors to forecast crude oil prices accurately, while this task is beset with difficulties and challenges due to the complex patterns in oil prices. This paper aims to propose a novel hybrid method to model and forecast the crude oil price by integrating median ensemble empirical mode decomposition (MEEMD) and mix-kernel extreme learning machine (Mix-KELM). Firstly, the emerging MEEMD is employed to decompose the crude oil price into several simple subseries. Secondly, a novel mix-kernel is developed for extreme learning machine (ELM) by combining the advantage of the local kernel (i.e., Radial Basis Function in learning ability) and global kernel (i.e., Sigmoid in generalization ability), with weights of the kernels optimized through genetic algorithm. Thirdly, the proposed Mix-KELM is applied to forecast the subseries of crude oil price, and the sub-forecasting results are integrated to generate the final results. The empirical results show that our proposed MEEMD-Mix-KELM model with different forecasting horizons significantly outperforms the benchmarks in terms of forecasting accuracy and robustness test. Taking one-step-ahead forecasting as an example, the proposed model exhibits the lowest prediction errors in terms of mean absolute error, symmetric mean absolute percentage error, and root mean squared error with values of 1.1767, 0.0135, and 1.5717, respectively.

1. Introduction

As a key input for modern production, crude oil continues to hold the largest share of the world's energy consumption (31%) in 2021 (BP, 2022). The substantial volume of oil consumption has drawn attention from various stakeholders, including governments, enterprises, and investors, who closely monitor oil price fluctuations due to their significant impact on energy security, production strategies, and portfolio management (Wang et al., 2017). To gain a deeper understanding of oil price fluctuations in advance, this paper, therefore, puts forward an innovative hybrid approach for prediction. Hybrid models, acknowledged for their capability to amalgamate the advantages of diverse single forecasting techniques, offer a more comprehensive and robust framework for improved prediction, particularly in discerning the hidden patterns of oil price data (Guan and Gong, 2023; Wu et al., 2023; Fang et al., 2023; Zhang et al., 2023).

The extreme learning machine (ELM) is a widely recognized artificial intelligence (AI) model utilized for effectively modeling intricate data patterns related to oil prices. In comparison to other neural networks (NN), ELM offers notable benefits in regard to computational efficiency and ease of implementation, as it employs a randomized approach to generate input layer weights and hidden layer biases (Li and Zhao, 2020). Despite its high performance, the inherent stochasticity of ELM can sometimes lead to fluctuations in its interpolating performance. To overcome this challenge, the kernel ELM (KELM) was proposed by imbedding kernel functions into the ELM model for feature mapping (Huang et al., 2012). Generally, kernels used in existing methods can be categorized into two groups: local kernels and global kernels (Peng et al., 2023). Local kernels such as radial basis function (RBF) tend to have strong interpolating (i.e., learning) capacity, while global kernels such as Sigmoid usually present good extrapolation (i.e., generalization) ability (Dong et al., 2020). To achieve optimal characteristics from both

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kernels, this paper attempts to develop a novel mix-kernel for ELM, combining the RBF kernel and Sigmoid kernel. Within this proposed mix-kernel framework, genetic algorithm (GA) is applied to fine-tune the weights allocated to the local and global kernels. GA, acknowledged as one of the most prevalent optimization methods, is selected for its inherent advantages, including global exploration, low algorithmic complexity, parallelism, heuristic nature, and strong adaptability to complex and dynamic problem domains (Lü et al., 2020; Zhou et al., 2022). Consequently, the novel mix-kernel extreme learning machine (Mix-KELM) is used to promote the accuracy of oil price prediction.

It is widely recognized that crude oil price data exhibits complex patterns arising from the interplay of various factors, including supply–demand dynamics (Liang et al., 2023), geopolitical shifts (Bouazizi et al., 2024), macroeconomic factors (Lyu et al., 2024), and financial indicators (Fiszeder et al., 2023). The complex patterns, particularly the nonlinear characteristics, present significant challenges to the task of oil price prediction. To reduce the complexity of oil prices, decomposition techniques are commonly utilized to decompose the price into several simple sub-series, which are subsequently analyzed or predicted individually (Xu and Wang, 2023; Salamai, 2023; Sun et al., 2022). For example, empirical mode decomposition (EMD) is currently one of the most powerful tools for time series decomposition, and has been proven effective for crude oil price analysis (Fang et al., 2023). However, the traditional EMD method frequently falls into mode mixing and mode splitting (Randall and Antoni, 2023), because real-world data typically exhibit both signal noise and intermittency. To address this issue, ensemble EMD (EEMD) is proposed by conducting EMD on an ensemble of the signals combined with standard normally distributed white noise. Though proven more effective, EEMD inevitably creates new mode splitting problems (Lang et al., 2020). To avoid this new problem, Lang et al. (2020) have recently introduced a novel variation of the EEMD method, namely median ensemble empirical mode decomposition (MEEMD), which mitigates the problem by employing the median operator to create ensembles of noisy intrinsic mode function (IMF) trials instead of the mean operator. We endeavor to assess the efficacy of this novel approach in enhancing the accuracy of oil price forecasting.

Overall, this paper aims to propose a novel hybrid method, by incorporating the newly proposed MEEMD method and the novel Mix-KELM method, thereby serving crude oil price forecasting. The contributions of this paper to the literature are twofold. (1) The emerging decomposition method, i.e., MEEMD, is introduced for the first time to decompose the crude oil price, to avoid pattern splitting, which may lead to inaccurate predictions. (2) A novel mix-kernel is developed for ELM to promote crude oil price forecasting, by integrating the best properties from both the global kernel (i.e., Sigmoid) and local kernel (i.e., RBF). To substantiate the superiority of our proposed MEEMD-Mix-KELM hybrid model, we conducted a comprehensive comparative analysis against individual models and other existing hybrid models.

The remaining part of the paper is structured as follows: the next section (i.e., Section 2) provides a comprehensive literature review on existing prediction techniques focusing specifically on the oil market. Section 3 introduces the methodological background underlying the hybrid model we constructed. Section 4 provides a detailed exposition of the newly proposed hybrid model. Data with detail descriptions is presented in Section 5. Experimental design is elaborated in Section 6. Empirical results accompanied by discussions are presented in Section 7. The final section of the paper presents an overview of the primary results and offers concluding remarks.

2. Literature review

Accurate crude oil price forecasting has always been a tough task, since it is driven by an array of complex factors. These factors, encompassing supply–demand fundamentals (Hao et al., 2020; Liang et al., 2023), geopolitical shifts (Zhang et al., 2022; Lyu et al., 2024), macroeconomic (Bouazizi et al., 2024; Zhang et al., 2024) and financial

indicators (Fiszeder, Faldziński, & Molnár, 2023; Zhang & Wang, 2019), have been extensively integrated into forecasting models to augment the precision of oil price predictions. Broadly, methods for forecasting crude oil prices are classified into three primary categories: econometric models, AI models, and hybrid models.

Early oil price forecast studies are mainly based on econometric models, among which autoregressive integrated moving average (ARIMA) (Xiang and Zhuang, 2013), generalized autoregressive conditional heteroskedasticity (GARCH) (Agnolucci, 2009; Mohammadi and Su, 2010), linear regression (LR) (Ye et al., 2006), random walk (RW) and vector error correct model (VECM) (Murat and Tokat, 2009) have been widely applied. Though they have a good linear fitting ability and short-term forecasting power, these kinds of models fail to characterize the nonlinear feature of crude oil price (Abdollahi and Ebrahimi, 2020).

AI models, in contrast, can well characterize the nonlinear and complexity of crude oil prices by their flexible functional form (Yuan et al., 2023). AI models like support vector Regression (SVR) (Yu et al., 2014; Wang et al., 2020; Karasu et al., 2020) and artificial neural network (ANN) are the most effective and popular tools for crude oil price forecasting. Furthermore, among the ANN models, ELM (Zhang et al., 2021; Li et al., 2022), back propagation neural network (BPNN) (Li et al., 2020), random vector functional link network (RVFL) (Tang et al., 2018), recurrent neural networks (RNN) (Guo et al., 2023), long short term memory (LSTM) (Li et al., 2021; Huang and Deng, 2021; Huang et al., 2023), bidirectional LSTM (Bi-LSTM) (Abdollahi, 2023; Xu et al., 2024), gate recurrent unit (GRU) (Jiang et al., 2022; Busari and Lim, 2021), and bidirectional GRU (Bi-GRU) (Fang et al., 2023) are widely utilized. Though proved powerful for forecasting, AI models are often sensitive to parameter settings, and may be prone to local optimization and overfitting issues (Zhang et al., 2021). For example, the prevalent ELM model is sensitive to the choice of kernel functions, particularly when using kernel-based variants such as the KELM.

Undoubtedly, various models, whether of econometric or AI origins, possess unique strengths in capturing different characteristics of crude oil prices. Combining these diverse strengths among models shows promise for enhancing predictions. Model averaging, which combines predictions from diverse models, has become a prevalent approach in oil price forecasting, with widely recognized methodologies such as Bayesian model averaging (BMA) and dynamic model averaging (DMA) (Filippidis et al., 2023; Naser, 2016; Drachal, 2021; Drachal, 2018). BMA assigns probabilities to models, while MDA assigns model weights to observations to estimate the best-performing model in a time-varying manner. Although model averaging largely improves the predictive performance of single models, it does not create a new model but aggregates existing models.

Emerging hybrid models tend to construct new models, by integrating the strengths of different models such as decomposition methods, optimization algorithms, and forecasting technologies. Thereinto, optimization algorithms such as GA (Abdollahi and Ebrahimi, 2020), particle swarm optimization (PSO) (Abdollahi, 2020; Sen and Choudhury, 2024), gorilla troops optimizer (GTO) (Li et al., 2022), artificial bee colony (ABC) (Wang et al., 2020), seagull optimization algorithm (SOA) (Jiang et al., 2022), and improved golden jackal optimization (IGJO) (Wu et al., 2023) can be used to optimize related parameters in AI models to enhance the forecasting ability. While decomposition methods, such as various EMD family models (Tang et al., 2018; Abdollahi, 2020; Wang et al., 2018; Li et al., 2022; Tang et al., 2020), wavelet decomposition (Lin et al., 2022), and variational mode decomposition (VMD) models (Zhang et al., 2023; Li et al., 2021; Bisoi et al., 2019), are widely utilized to decompose original crude oil prices into simpler sub-series using the “divide and conquer” principle (Sun et al., 2018). It's worth noting that the prevalent EMD and EEMD models frequently encounter mode splitting issues (Lang et al., 2020), prompting the selection of the MEEMD model for oil price decomposition in this study.

Building upon existing literature, this paper aims to improve the

West Texas Intermediate (WTI) crude oil price prediction by proposing a novel hybrid approach, namely MEEMD-Mix-KELM. In this hybrid model, an emerging and effective decomposition method i.e., MEEMD is applied to decompose the WTI oil price into several simple and meaningful subseries. Afterward, a Mix-KELM method is proposed by constructing a novel mix-kernel for ELM to forecast each subseries of crude oil price. And finally, ultimate forecasted values of crude oil prices are obtained by combining the sub-prediction results. The proposed hybrid model addresses two weaknesses of existing research: the problem of mode splitting during data decomposition the limitations arising from the ELM model's use of a singular kernel function. In particular, utilizing the novel decomposition model MEEMD for oil price decomposition effectively mitigates the issue of mode splitting, encountered in existing decomposition models. Furthermore, the proposed Mix-KELM method can significantly enhance the level of oil price prediction compared to singular global or local kernel functions.

3. Background and related methodology

In this section, background and description of the related methods that used to construct the proposed hybrid model framework, namely the MEEMD method, ELM models (ELM, KELM, and Mix-KELM), and GA, are introduced in particular.

3.1. Median ensemble empirical mode decomposition

EMD, first proposed by Huang et al. (1998), is presently considered to be one of the most effective techniques for performing time–frequency analysis, and has been successfully applied to economic variable forecasting, especially for the crude oil price. By using this method, it becomes possible to adaptively decompose the oil price into a collection of modes (referred to as IMFs and residue), each of which represents the fluctuation of oil prices at a specific time scale. Following the “divide and conquer” principle, each mode can be forecasted individually and combined to generate the final forecasted results.

Nevertheless, mode mixing and mode splitting are bound to incur in the EMD process, because the crude oil markets, being complex systems, often exhibit noise and signal intermittency. To overcome this problem, Wu and Huang (2009) developed a noise-assisted data decomposition method called EEMD by performing EMD on an ensemble of the signal plus white Gaussian noise. According to the statistical characteristics of the Gaussian distribution, the EEMD method has been shown to effectively mitigate the issue of mode mixing in the EMD process, which arises due to the presence of noise and intermittency in the IMFs (He and Wang, 2021).

Despite the widespread practical applications of the EEMD technique (Tang et al., 2018; Zhang et al., 2021), it is inevitable that new mode splitting will occur due to two primary reasons. First, the EMD equivalent filter's overlapping region may lead to mode splitting in signals of a certain scale (Wu and Huang, 2009). Second, it is worth noting that the introduction of Gaussian white noise may not ensure uniformity across all scales, which could potentially lead to unexpected fluctuations and irregularities in the signal (Xie et al., 2016). To mitigate the issue of additional mode splitting, Lang et al. (2020) proposed an enhanced version of the EEMD algorithm named MEEMD, by taking the median operator instead of the mean operator, to ensemble the noisy IMF trials.

Given oil price data $x(t)$, $t = 1, \dots, T$, and a predefined noise amplitude ε , the newly proposed MEEMD is performed through three steps:

(1) Create the ensemble $y_n(t) = x(t) + \varepsilon w_n(t)$ for $n = 1, \dots, N$, where $w_n(t) \sim N(0, 1)$ denotes the white Gaussian noise, and N is the number of white Gaussian noise added to the crude oil price.

(2) Decompose each member of $y_n(t)$ into M_n modes with the standard EMD, to obtain the mode sets, i.e., IMFs $\{d_m^n\}_{m=1}^{M_n-1}$ and residues $r_n(t)$, where

$$y_n(t) = \{d_m^n\}_{m=1}^{M_n-1} + r_n(t) \quad (1)$$

(3) Use the medium operator to calculate final modes, with IMFs and residue computed as

$$d_m(t) = \text{median}\{d_m^1(t), d_m^2(t), \dots, d_m^N(t)\} \quad (2)$$

$$r(t) = \text{median}\{r_1(t), r_2(t), \dots, r_N(t)\} \quad (3)$$

where the medium operator is defined as

$$\text{Mode}^{\text{Median}}(t) = \frac{\text{Mode}^{\lfloor (N+1)/2 \rfloor}(t) + \text{Mode}^{\lceil (N+1)/2 \rceil}(t)}{2} \quad (4)$$

in which $\text{Mode}^n(t)$, $n = 1, 2, \dots, N$, represents the ordered mode list at time t , which is derived from N -independent realizations of EMD-based decomposition.

3.2. Extreme learning machine models

In this section, we provide an overview of the fundamental principles underlying ELM and KELM, which serve as the basis for our proposed Mix-KELM method.

3.2.1. Extreme learning machine

ELM, first introduced by Huang et al., 2006, has since been recognized as a powerful tool for classification and forecasting. Compared with traditional learning schemes, ELM offers several benefits, including scalability, lower computational complexity, simple implementation, and high learning ability (Fossaceca et al., 2015).

Assuming that the training set is represented as $\{(x_i, t_i) | x_i \in \mathbb{R}^d, t_i \in \mathbb{R}^m\}_{i=1}^N$, where the number of input neurons and output neurons are denoted as d and m respectively, with N being the number of samples. For the i -th instance, x_i represents the corresponding feature information, while t_i denotes label information related to the crude oil price. For the regression case in this paper, $m = 1$. The output function of an ELM network comprising L hidden neurons can be expressed as follows:

$$f_{elm}(x) = \sum_{i=1}^L \beta_i g(\omega_i x_i + b_i), j = 1, 2, \dots, N \quad (5)$$

where β_i denotes the weight matrix connecting the hidden neurons to the output neurons, $g(\bullet)$ represents the activation function, ω_i is the randomly assigned weight vector that connects the input nodes and hidden neurons, and b_i represents the bias of hidden layer neurons. Eq. (5) can be simply redefined as

$$H\beta = T \quad (6)$$

$$H = (\omega_1, \dots, \omega_L, b_1, \dots, b_L, x_1, \dots, x_N) \\ = \begin{bmatrix} g(\omega_1 x_1 + b_1) & \dots & g(\omega_L x_1 + b_L) \\ \vdots & \ddots & \vdots \\ g(\omega_1 x_N + b_1) & \dots & g(\omega_L x_N + b_L) \end{bmatrix} \quad (7)$$

$$\beta = (\beta_1, \beta_2, \dots, \beta_L)^T \quad (8)$$

$$T = (T_1, T_2, \dots, T_N)^T \quad (9)$$

where H refers to the output matrix of the hidden layer, and T denotes the ground truth label matrix for the training set. In the ELM technique, the input weights, denoted as ω_i and the hidden bias, denoted as b_i , are randomly assigned without the need for a tuning process (Huang et al., 2012). Thereby, Eq. (6) can be considered as a linear system and its solution is

$$\beta = H^\dagger T \quad (10)$$

where $H^\dagger$ is the pseudo-inverse of the matrix H and is computed using

the Orthogonal projection method as $H^\dagger = H^T(HH^T)^{-1}$.

To get a better predictive ability and a stable output solution, a positive number known as regularization coefficient C is incorporated into the diagonal of HH^T . Hence, Eq. (10) can be redefined as

$$\beta = H^T(HH^T + \frac{I}{C})^{-1}T \quad (11)$$

The primary goal of ELM is to minimize both the training error and the output weight. To achieve this objective, the original ELM formulates the objective function as follows:

$$\min_{\beta} \frac{1}{2} \|\beta^2\| + \frac{C}{2} \|H\beta - T\|^2 \quad (12)$$

Based on the convex optimization theory, the training of ELM can be formulated as solving the following dual optimization problem:

$$L_{KELM} = \frac{1}{2} \|\beta\|^2 + \frac{C}{2} \|\varepsilon\|^2 + \text{Tr} \alpha^T (T - H\beta - \varepsilon) \quad (13)$$

where ε denotes the error vector, and α denotes the Lagrangian multiplier. The optimal solution of the dual optimization problem in ELM, as per the KKT theorem, leads to the expression of the output function as:

$$f(x) = h(x)H^T \left(\frac{I}{C} + HH^T \right)^{-1} T \quad (14)$$

3.2.2. Kernel extreme learning machine

Compared to traditional ELM methods, KELM, proposed by Huang et al., 2012, adopts the kernel transformation method instead of randomly assigning input parameters. This method overcomes the difficulty of choosing the appropriate number of hidden nodes and has demonstrated improved generalization performance. The ELM kernel matrix can be expressed as:

$$K_{ELM} = HH^T \quad (15)$$

Hence, the output function in Eq. (14) can be rewritten as:

$$f(x) = h(x)H^T(K + \frac{I}{C})^{-1}T = [k(x, x_1); \dots; k(x, x_N)](K + \frac{I}{C})^{-1}T \quad (16)$$

The existing ELM is inherently non-deterministic and kernel-dependent, which necessitates careful consideration of kernel selection. To address this issue, diverse kernels are designed to improve the generalization performance of the original ELM. Popular kernels are RBF kernel, Sigmoid kernel, linear kernel, polynomial kernels, with their respective formulas displayed in Table 1.

3.2.3. Mix-kernel extreme learning machine

Considering the significant influence of kernel functions on the performance of ELM, the selection of an optimal kernel has become a major research focus among scholars. Generally, there are two categories of kernel functions available for ELM, namely the global kernel and the local kernel. Each category has its strengths. Accordingly, the local kernel function (e.g., RBF) exhibits superior interpolating characteristics, i.e., the ability to fit the training data. While the global kernel function (e.g., Sigmoid) possesses remarkable extrapolation capabilities, i.e., the ability to predict data outside the range of the training sets. Single kernel function may have certain limitations in data classification

and prediction. Thus, this paper aims to propose a new mix-kernel function for ELM by integrating both local (RBF) and global (Sigmoid) kernel functions. The mix-kernel is designed to be a linear combination of Sigmoid and RBF kernel, i.e.,

$$K_{Mix} = \omega K_{RBF} + (1 - \omega) K_{Sig}, (0 < \omega < 1) \quad (17)$$

where K_{RBF} and K_{Sig} are kernel functions of RBF and Sigmoid, and their weights are ω and $1 - \omega$, respectively. The value of ω is optimized by GA in this study. By integrating the complementary features of these two kernel functions, our mixed kernel is expected to yield improved predictive performance compared to single kernel approaches. In other words, the proposed Mix-KELM is expected to beat both RBF integrated KELM (RBF-KELM) and Sigmoid integrated KELM (SIG-KELM).

3.3. Genetic algorithm

GA, proposed by Holland (1975) based on the biological evolution of living organisms, has been shown to be an effective metaheuristic optimization algorithm for resolving a wide range of optimization problems. In view of the parameter sensitivity of AI models, GA is widely applied to parameter optimization. This study, therefore, introduces GA to identify the optimal set of parameters for the proposed Mix-KELM model, including those in RBF kernel (i.e., σ), Sigmoid kernel (i.e., θ and γ), the mix-kernel (i.e., ω) and ELM (i.e., regularization coefficient C).

The evolution theory of GA is to evaluate the fitness function of every individual in a population to determine which individuals can enter the next generation. To achieve this goal, three operators are performed, including selection, crossover and mutation.

(1) **Selection** aims to select the elite group of generation. To achieve this, a fitness function, such as root mean square error (RMSE), can be applied to evaluate the performance of each individual within the population. Only individuals with optimal fitness values are selected to be retained, while those with suboptimal fitness values are discarded. The selection process can be expressed as follows

$$G_i(p\{x_0, \dots, x_b\}) = \begin{bmatrix} p_0 \\ \vdots \\ p_n \end{bmatrix} = \begin{bmatrix} x_{00} & \dots & x_{0b} \\ \vdots & \ddots & \vdots \\ x_{n0} & \dots & x_{nb} \end{bmatrix} \quad (18)$$

where i denotes the number of iterations, p represents the number of individuals in the current generation, b is the total population in the current generation, and n refers to the number of parameters in each individual.

(2) **Crossover** aims to generate new individuals. The randomly selected individuals are specified as parents and they can generate a pair of offspring by swapping part of their gene information. The offspring are retained for the next generation.

(3) **Mutation** aims to randomly modify the parents' gene information. In order to get diverse offspring, this operator is performed by introducing new patterns into parents' gene information.

4. The proposed hybrid model

A novel hybrid method, combining multi-scale data decomposition (i.e., MEEMD) and mixed kernel function development for ELM (i.e., Mix-KELM), is proposed in this work to improve oil price prediction accuracy. Consequently, the proposed method involves three primary steps, i.e., oil price decomposition, mix-kernel design, and oil price forecasting, as shown in the general framework depicted in Fig. 1.

Step 1: Oil price decomposition

Taking inspiration from the principle of "divide and conquer," this step involves applying an emerging decomposition method called MEEMD to decompose the oil price into n easily scrutable subseries, including $(n - 1)$ IMFs and a residue term.

Step 2: Mix-kernel design

The choice of kernel function is widely recognized as a critical factor

Table 1
Popular kernel functions for ELM.

Kernels	Formulas
RBF	$K(x_1, x_2) = \exp\left(-\frac{\ x_1 - x_2\ ^2}{2\sigma^2}\right)$
Sigmoid	$K(x_1, x_2) = \tanh(\theta x_1^T x_2 + \gamma)$
Linear	$K(x_1, x_2) = x_1^T x_2$
Polynomial	$K(x_1, x_2) = (x_1^T x_2 + 1)^d$

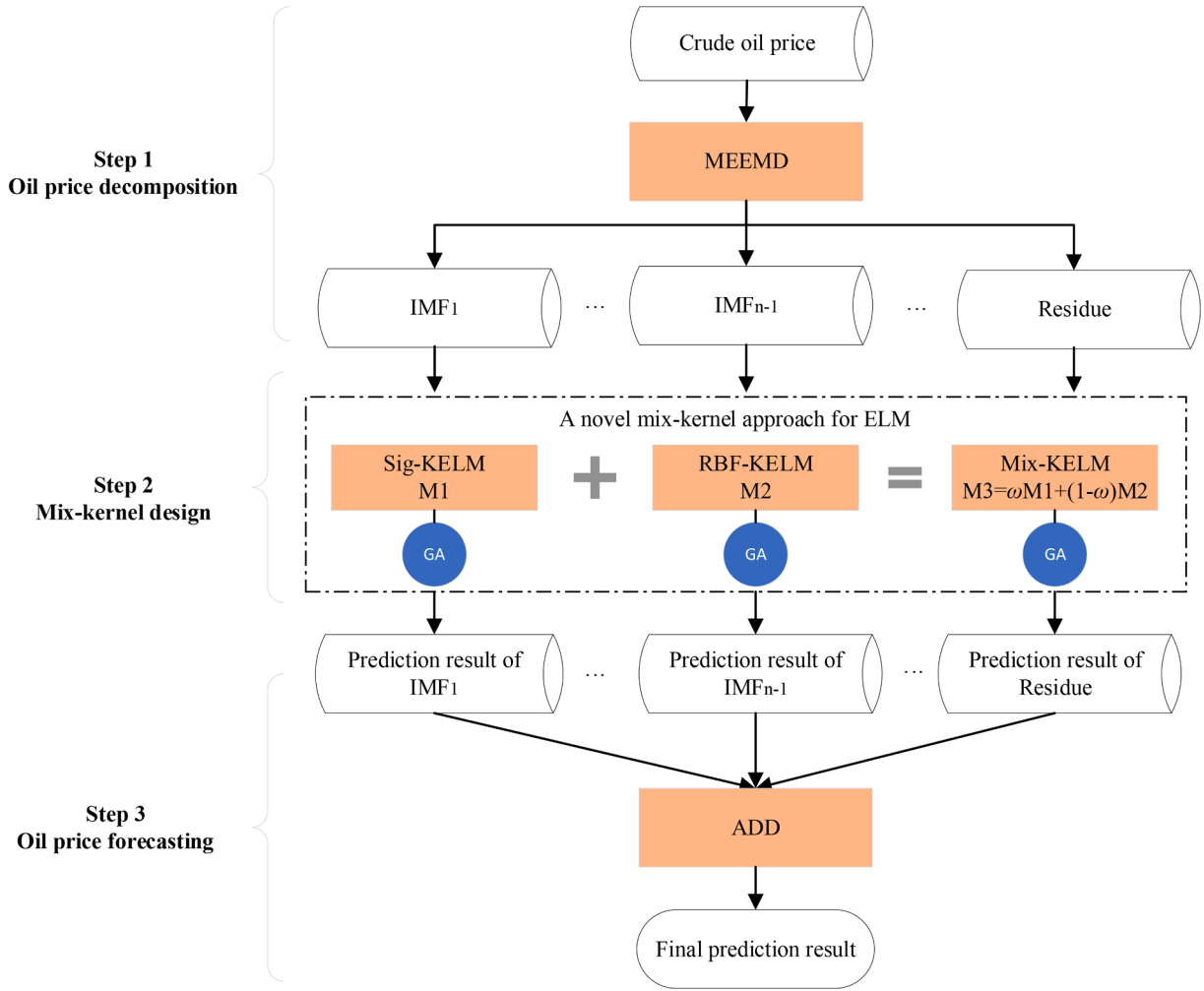


Fig. 1. General framework for the proposed hybrid model.

affecting the performance of ELM. Therefore, in this step, a novel mix-kernel function that integrates the advantages of two existing kernel functions is proposed for ELM. The mix-kernel used in this study is a composite kernel that consists of an RBF kernel and a Sigmoid kernel, whose weights are set to sum up to 1. To enhance the generalization capability of the proposed mix-kernel ELM (i.e., Mix-KELM), GA is employed in this paper to optimize the parameters in Mix-KELM, including those in RBF kernel (i.e., σ), Sigmoid kernel (i.e., θ and γ), the mix-kernel (i.e., ω) and ELM (i.e., regularization coefficient C).

Step 3: Oil price forecasting

In this step, the proposed Mix-KELM is utilized to forecast the n subseries of oil price. For each individual prediction, the m -step-ahead forecast can be generated by: $\hat{X}(t+m) = f(X(t), X(t-1), \dots, X(t-p))$, where $\hat{X}(t+m)$ denotes the m -step-ahead forecasting value of each subseries at time t , $X(t)$ represents the actual oil price value at time t , and p is the lag order. Final predictions for the crude oil price can be obtained by combining the forecasting outcomes of all subseries.

According to the above steps, the pseudocode of MEEMD-Mix-KELM is displayed as follows:

Algorithm 1 MEEMD-Mix-KELM

Input: Oil price data $x(t), t = 1, \dots, T$
 Output: Final predictions for the oil price
 1: Create the ensemble $y_n(t) = x(t) + \varepsilon w_n(t)$ for $n = 1, \dots, N$
 2: Decompose each member of $y_n(t)$ into M_n modes with the standard EMD, to obtain the sets, i.e., IMFs $\{d_m^i\}_{m=1}^{M_n-1}$ and residues $r_n(t)$

(continued on next column)

(continued)

Algorithm 1 MEEMD-Mix-KELM

- 3: Use the medium operator to calculate final modes, with IMFs and residue computed $as d_m(t) = median\{d_m^1(t), d_m^2(t), \dots, d_m^N(t)\}$ and $r(t) = median\{r_1(t), r_2(t), \dots, r_N(t)\}$
- 4: Calculate the mix-kernel matrix of IMF_n , $K_{mix} = \omega K_{RBF} + (1-\omega)K_{Sig}$, ($0 < \omega < 1$), where $K_{RBF}(x_1, x_2) = \exp(-\|x_1 - x_2\|^2 / 2\sigma^2)$, $K_{Sig}(x_1, x_2) = \tanh(vx_1^T x_2 + \gamma)$
- 5: Use GA to identify the optimal set of parameters for the proposed Mix-KELM model, three operators are performed, including selection, crossover and mutation
- 6: Calculate the output function $f(x) = h(x)H^T(K_{mix} + I/C)^{-1}T = [K_{mix}(x, x_1); \dots; k(x, x_N)](K_{mix} + I/C)^{-1}T$, i.e., Prediction results of IMFs $d_m(t)$ and residue $r(t)$
- 7: Calculate final predictions for the crude oil price by combining all of the prediction results of IMFs $d_m(t)$ and residue $r(t)$

5. Data with descriptions

The study utilizes the WTI crude oil spot price as the data sample, as shown in Fig. 2 and described in Table 2, which was collected from the U.S. Energy Information Administration (<https://www.eia.doe.gov>). The data comprised 877 weekly observations, ranging from January 5, 2007 to October 20, 2023. The average and median WTI crude oil prices for the period were \$72.79 and \$71.80, respectively, with a standard deviation of \$22.84. The heavy volatility of the WTI crude oil price highlights its significance in market forecasting. The time series plot in Fig. 2 shows that the highest price peak of \$142.46 occurred on July 4,

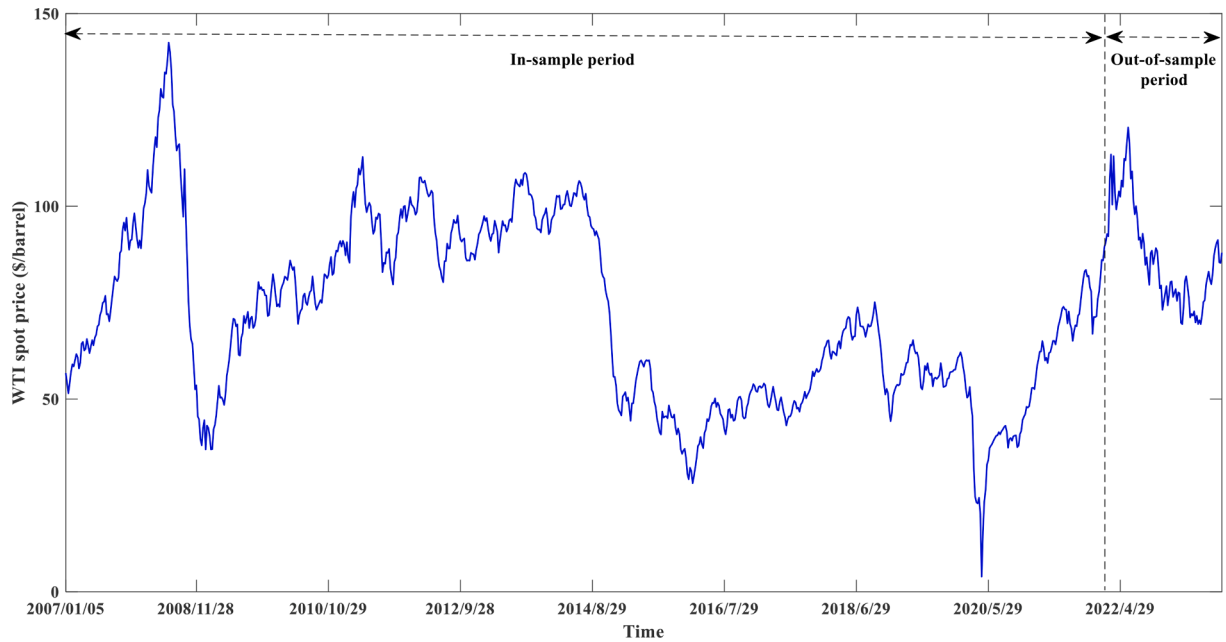


Fig. 2. Movements of WTI crude oil price.

Table 2
Statistics for WTI crude oil price.

	Full sample	In-sample	Out-of-sample
Mean	72.79	71.19	87.10
Median	71.80	68.85	84.41
Maximum	142.47	142.46	120.44
Minimum	3.92	3.92	69.42
Std. Dev.	22.84	23.13	13.27
Skewness	0.15	0.26	0.71
Kurtosis	2.41	2.42	2.55
Jarque-Bera	16.30	20.09	8.10
Probability	0.00	0.00	0.02
Observations	877	789	88

2008 (observation 79), while the lowest price of \$3.92 occurred on April 24, 2020 (observation 695).

According to Abdollahi and Ebrahimi (2020), the sample series was partitioned into two distinct sets: a training dataset encompassing the initial 90% of the sample period and a testing dataset comprising the remaining 10%. Thus, the in-sample period ranges from January 5, 2007 to February 11, 2022, and the out-of-sample period ranges from February 18, 2022 to October 20, 2023. The performance of the proposed framework was evaluated using multi-step ahead predictions at different horizons ranging from one to three weeks, which provides a comprehensive assessment of its robustness.

6. Experimental design

This section expounds upon the experimental design of our work, encompassing an elaboration of the evaluation criteria employed to assess the forecasting performance, and a detailed delineation of the parameters specified in the various forecast models.

6.1. Evaluation criteria

To assess the precision of different forecasting models, three widely used criteria are chosen, namely, mean absolute error (MAE), symmetric mean absolute percentage error (SMAPE), and root mean squared error (RMSE), as presented below:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (18)$$

$$\text{SMAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{(\hat{y}_i + y_i)/2} \quad (19)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (20)$$

where $\hat{y}_t$ denotes the predicted value of oil price and y_t is the true value at time t , respectively. The data size of the testing data is represented as N .

To undertake a comprehensive comparative analysis of diverse forecasting models from a statistical viewpoint, this paper employs the Diebold–Mariano (DM) test (Diebold and Mariano, 2002) and the model confidence set (MCS) test (Hansen et al., 2011) to examine the significance of prediction disparities. The loss function utilized in our study is RMSE. In the DM test, the null hypothesis assumes that the RMSE of the tested model A is not significantly worse than that of the benchmark model B. The DM statistic can be formally defined as follows:

$$S = \frac{\bar{g}}{(\hat{V}/N)^{1/2}} \quad (21)$$

where $\bar{g} = \frac{1}{N} \sum_{t=1}^N [(y_t - \hat{y}_{A,t})^2 - (y_t - \hat{y}_{B,t})^2]$, $\hat{V} = \gamma_0 + 2 \sum_{l=1}^{\infty} \gamma_l$, ($\gamma_l = \text{cov}(g_t, g_{t-l})$), and $\hat{y}_{A,t}$ and $\hat{y}_{B,t}$ represent the forecasting outcomes of models A and B, respectively, at time t .

The MCS test doesn't require setting a benchmark model; instead, it involves comparing multiple models through equivalence tests and elimination rules. Subsequently, the optimal model set can be determined within a specified confidence level. The MCS test, combined with the Bootstrap method, calculates two statistics: TR and TMAX, which are calculated as:

$$\text{TR} = \max_{i,j \in \mathcal{M}} \frac{|\bar{d}_{ij}|}{\sqrt{\widehat{\text{Var}}(d_{ij})}} \quad (22)$$

$$TMAX = \max_{i \in \mathcal{W}} \frac{|\bar{d}_{i..}|}{\sqrt{\widehat{\text{Var}}(\bar{d}_{i..})}} \quad (23)$$

where $\bar{d}_{ij} = \bar{g}, \bar{d}_{i..} = \frac{1}{M} \sum_{j \in \mathcal{W}} \bar{d}_{ij}$, and $\widehat{\text{Var}}$ denotes the variance operator.

6.2. Model specification

As for forecasting technologies, seven prediction methods, i.e., RW, ARIMA, BPNN, SVR, KELM with single kernels (i.e., RBF-KELM and SIG-KELM) and mix-kernel (Mix-KELM) are utilized in this paper. With regard to the decomposition models, three decomposition models from the EMD family, namely EMD, EEMD and MEEMD, are taken into consideration to improve the forecasting accuracy.

Multi-step forecasting approaches were employed for the subseries of the WTI crude oil price. In ARIMA ($p-d-q$), parameter values were determined by minimizing the Akaike information criterion (AIC). As for BPNN, a conventional three-layer neural network architecture with two hidden nodes is devised, and the model is trained iteratively for 500 epochs using the gradient descent method with a fixed learning rate of 0.01. For SVR, RBF is selected as the kernel function, and the value of parameter σ^2 is optimized by GA. In KELM methods with single kernels, the Sigmoid and RBF functions are chosen, and their parameters are also optimized by GA. In terms of the proposed Mix-KELM, the weight and parameters in the single kernel are detected by GA. In GA, a population size of 20, a maximum number of iterations of 50, and the operators of normGeomSelect (for selection), arithXover (for crossover), and non-UnifMutation (for mutation), are used, respectively. Additionally, to ensure the robustness of the models, the present study conducts one-step, two-step, and three-step ahead forecasting analyses for all proposed methods. Finally, the lag order of the AI techniques is determined as 4 using the trial and error method, based on the highest level of fitting accuracy observed among the training samples.

7. Experimental results

Based on the general framework depicted in Fig. 1, this study consists of three essential steps, i.e., oil price decomposition, mix-kernel design and oil price forecasting. The three key steps yield relevant results as expounded in Sections 7.1-7.3. Additionally, the discussion in Section 7.4 delves into the prospective application of the proposed mixed-kernel function in SVR.

7.1. Decomposition results

In this study, the emerging effective decomposition tool, MEEMD, is employed to obtain subseries of crude oil prices. The resulting subseries are presented in Fig. 3 in order of increasing timescale, and their corresponding average periods, measured via Fourier transform, are listed in Table 3. The smallest time scale (IMF 1, average period is 3.6 weeks) corresponds to random fluctuations in crude oil price, while the largest time scale (residue, average period is 877 weeks) captures the long-term tendencies intrinsic to the underlying crude oil price. These observations highlight the coexistence of both short-term stochastic fluctuations and long-term tendencies in crude oil prices, which can be effectively separated using the MEEMD technique. Importantly, it can be concluded from Fig. 3 and Table 3 that the subseries are much simpler and more periodic compared to the original crude oil price. Therefore, forecasting on the simpler subseries may be easier and more accurate than directly

Table 3

Timescales (average period) of the subseries obtained by MEEMD.

Mode	Average period (weeks)	Mode	Average period (weeks)
IMF 1	3.6	IMF 4	46.2
IMF 2	8.1	IMF 5	73.1
IMF 3	17.9	Residue	877.0

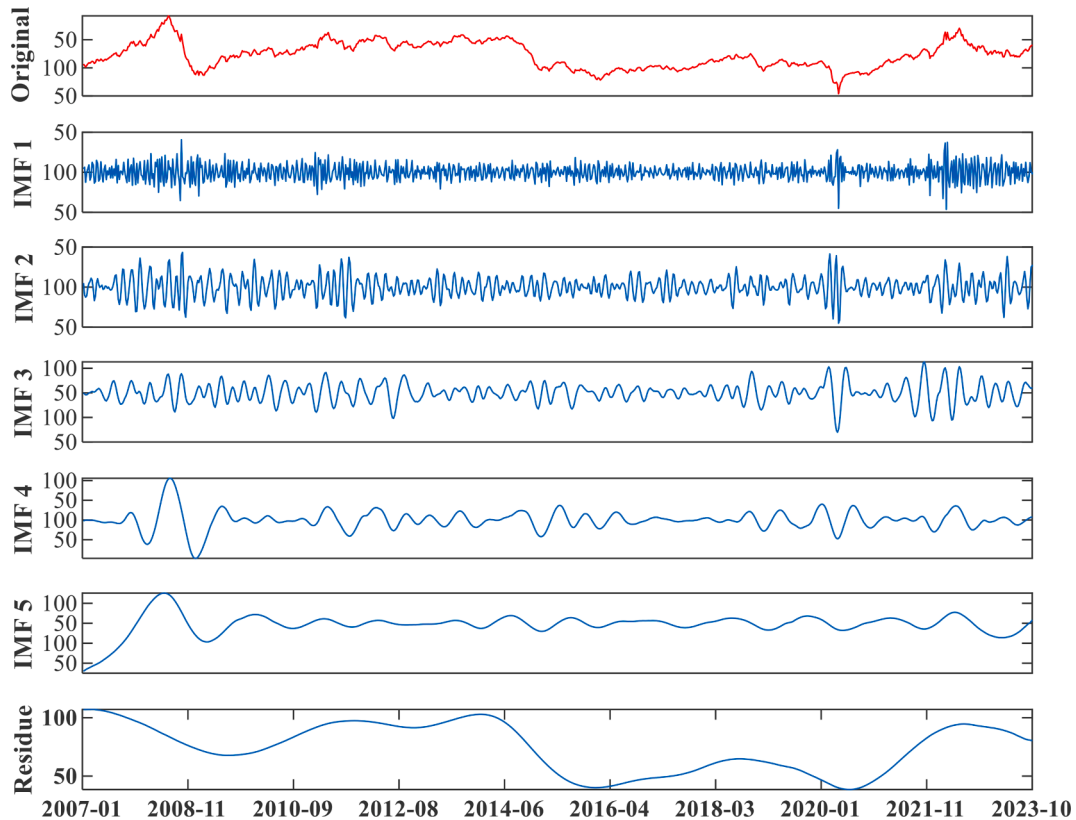


Fig. 3. Decomposition outcomes of international crude oil price using the MEEMD approach.

forecasting on the complex original crude oil price.

7.2. Mix-kernel results

To perform individual forecasts for each subseries of crude oil price, the mix-kernel is designed to fit ELM, by combining the advantages of both global kernel (i.e., Sigmoid) and local kernel (i.e., RBF).

Using $x_i = 0.2$ as the testing point, the value movement of the RBF kernel and Sigmoid kernel with varying parameters are depicted in Fig. 4. With regards to the RBF kernel, the kernel value reaches a peak near the test point but tends to decrease for data points further away from it. Besides, it has been observed that the local characteristics of the RBF kernel decrease as the value of the parameter σ increases. Such local characteristics are linked to the kernel's interpolating ability, where a strong local characteristic results in a more robust interpolating ability. In contrast, the Sigmoid kernel comprises two parameters in its kernel function (as described in Table 1), where γ is set to 1 for ease of explanation. The kernel value of Sigmoid varies with the turning of the θ , and unlike the RBF kernel, the Sigmoid kernel demonstrates apparent global characteristics. Notably, every data point in the set x significantly influences the kernel value of the test point $x_i = 0.2$, regardless of its actual distance from the test point. As the parameter θ decreases, the global characteristics of the Sigmoid kernel function become more pronounced.

In practical applications, the GA serves as a powerful optimization tool to obtain optimal parameters that ensure an accurate generation process of the mix-kernel. The crucial parameters that require optimization comprise the σ in the RBF kernel, the γ and θ in the Sigmoid kernel and finally the weight ω in the mix-kernel. For each subseries of the crude oil price, the parameters undergo individual optimization to conform to their data characteristics. For the test point $x_i = 0.2$, the mix-kernel with varying weights (of RBF kernel and Sigmoid kernel) is depicted in Fig. 5. It is obvious that the mix-kernel demonstrates both local and global behavior. An increase in the weight of the RBF kernel function intensifies the local characteristics of the mix-kernel function while diminishing the global characteristics. Consequently, we believe that the proposed mix-kernel has amalgamated the advantage of the local kernel (i.e., excellent interpolating capacity) and global kernel (i.e., excellent extrapolation ability).

The proposed mix-kernel with regenerate properties, is applied to fit the ELM model, thus forming a novel hybrid model, i.e., Mix-KELM. The novel hybrid model is finally employed to forecast the subseries of crude oil price, which is obtained from MEEMD.

7.3. Forecasting results

This experiment aims to demonstrate the superiority of our proposed MEEMD-Mix-KELM in crude oil price forecasting. To achieve this

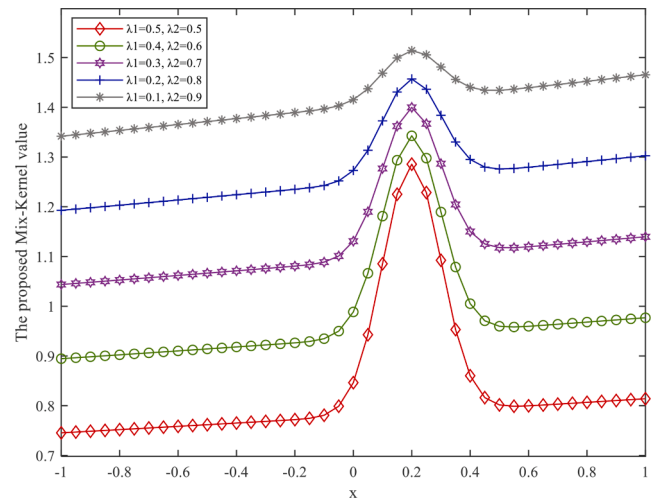


Fig. 5. Values of the mix-kernel with different weights, with fixed parameters $\sigma = 0.1$, $\theta = 0.1$, $\gamma = 1$.

objective, three groups of comparative experiments are designed, respectively. First, four single benchmarking models, namely RW, ARIMA, BPNN, and SVR, are taken into consideration to validate the strength of KELM in enhancing oil price forecasting. Second, as benchmark models for the proposed Mix-KELM, the study employs KELM models with a local kernel (RBF) and a global kernel (Sigmoid), which are contrasted against the proposed approach that uses a combination of both kernels. Third, based on the optimal single forecasting model (i.e., Mix-KELM model), two hybrid benchmarking models with different decomposition methods (i.e., EMD and EEMD) are constructed for comparison, to verify the advantage of MEEMD. By conducting the three-step comparison, the strength of the proposed model is thoroughly validated.

Tables 4-6 show the detailed calculation results of various forecasting techniques for WTI crude oil price across different horizons using MAE, SMAPE, and RMSE criteria. These tables furnish insightful deductions that can be drawn from the observed outcomes.

First, as for the single forecasting technologies shown in Table 4, KELM outperforms other econometric mode (i.e., RW, ARIMA) and AI models (i.e., BPNN and SVR) in oil price forecasting, with the minimum error criteria (i.e., MAE, SMAPE and RMSE) at all horizons from 1 to 3. In this experimental group, the same kernel (i.e., RBF) is chosen for both SVR and KELM, and observed that they generally provide better prediction accuracy than other models such as RW, ARIMA and BPNN. However, KELM still performs better than SVR in our experiment, leading to its selection as the preferred forecasting technology in this study. It is unsurprising that KELM outperforms the other three models

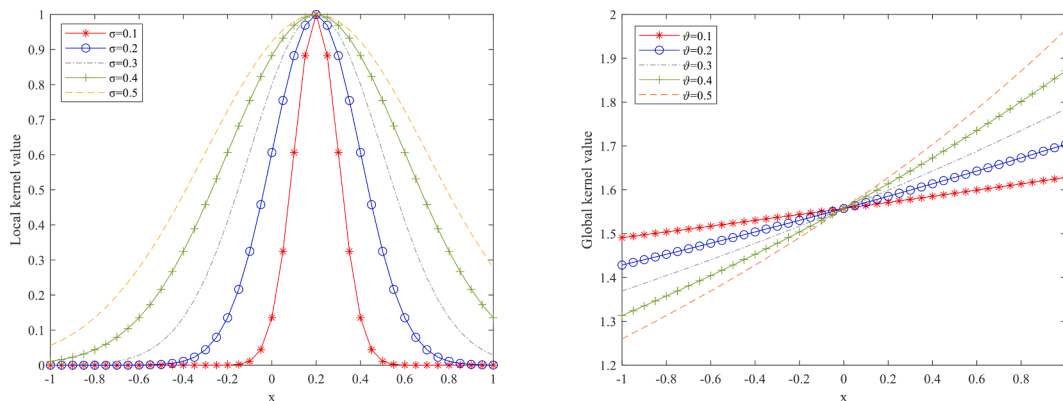


Fig. 4. Values of local kernel (i.e., RBF kernel) function (left) and global kernel (i.e., Sigmoid kernel) function.

Table 4

Forecasting performance comparison of different single models.

Models	One-step-ahead			Two-step-ahead			Three-step-ahead		
	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE
RW	3.9134	0.0440	5.1319	5.0482	0.0572	6.4389	5.9375	0.0668	7.5073
BPNN	3.5287	0.0396	4.6170	4.8873	0.0559	6.0759	5.6398	0.0640	6.9452
ARIMA	3.4971	0.0392	4.6106	4.6793	0.0532	6.0884	5.4661	0.0618	6.9451
SVR	3.4561	0.0388	4.5014	4.6458	0.0530	5.9336	5.5316	0.0627	6.7832
KELM	3.3838	0.0381	4.3900	4.4947	0.0508	5.7636	5.4268	0.0615	6.7150

Table 5

Forecasting performance comparison of KELM models with single kernel and mix-kernel.

Models	One-step-ahead			Two-step-ahead			Three-step-ahead		
	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE
RBF-KELM	3.3838	0.0381	4.3900	4.4947	0.0508	5.7636	5.4268	0.0615	6.7150
SIG-KELM	3.2987	0.0377	4.3770	4.6814	0.0533	5.8530	5.3014	0.0602	6.6386
Mix-KELM	3.1133	0.0354	4.1876	4.3764	0.0498	5.5200	5.1358	0.0581	6.3481

Table 6

Forecasting performance comparison of different EMD family-based hybrid prediction models.

Models	One-step-ahead			Two-step-ahead			Three-step-ahead		
	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE	MAE	SMAPE	RMSE
EMD-Mix-KELM	2.0283	0.0231	2.5545	2.1736	0.0244	2.8300	2.8099	0.0321	3.3787
EEMD-Mix-KELM	1.5222	0.0171	1.9626	2.0847	0.0235	2.6249	2.3077	0.0260	2.9924
MEEMD-Mix-KELM	1.1767	0.0135	1.5717	1.8862	0.0213	2.4358	2.1437	0.0243	2.6732

in oil price forecasting, consistent with findings from [Sun, Zhao, & Sun, 2022](#); [Wang, Athanasopoulos, Hyndman, & Wang, 2018](#); [Wu, Wu, & Zhu, 2019](#) and [Li et al. \(2022\)](#). This is attributed to its efficiency in handling extensive datasets, streamlined operational processes with minimal parameter tuning ([Alaba et al., 2019](#)).

Second, our proposed Mix-KELM outperforms KELM with single kernels in terms of all errors at all horizons, as shown in [Table 5](#). For instance, for horizon = 1, the mixed-kernel-based ELM (i.e., Mix-KELM) reduces MAE by 5.62%, SMAPE by 6.10%, and RMSE by 4.33% compared to the global kernel-based ELM (i.e., SIG-KELM). Further, compared to the local kernel-based ELM (i.e., RBF-KELM), the mixed type (i.e., Mix-KELM) exhibits greater improvements, with MAE reducing by 7.99%, SMAPE by 7.09%, and RMSE by 4.61%, respectively. The results suggest that the proposed Mix-KELM model, which combines the strengths of both local (in interpolating) and global (in extrapolating) kernels, outperforms other models and may be a superior choice for forecasting the crude oil price.

Third, based on the optimal Mix-KELM selected by the first and second group comparison, it is obvious that the prediction performances of the decomposition-ensemble models outperform those without. As illustrated in [Table 6](#), the forecasting errors of EMD family-based Mix-KELM models (i.e., EMD-Mix-KELM, EEMD-Mix-KELM and MEEMD-Mix-KELM) are all lower than single forecasting model. For example, we find the RMSE of Mix-KELM are between 4.1876 (at one-step-ahead forecasting) and 6.3481 (at three-step-ahead forecasting), while the RMSE of decomposition-ensemble models are between 1.5717 (in MEEMD-Mix-KELM at one-step-ahead forecasting) and 3.3787 (in EMD-Mix-KELM at three-step-ahead forecasting). The main reason lies in the advantage of decomposition-ensemble models to simplify complex problems, mitigate modeling complexities, and enhance predictive precision.

Fourth, focusing on the EMD family models, MEEMD outperforms the other two counterparts (i.e., EMD and EEMD) in promoting the Mix-KELM. The result is tenable for all error criteria from one-step-ahead to three-step-ahead. It is not surprising that the EEMD can defeat EMD, since EEMD can effectively mitigate the issue of mode mixing, caused by the presence of noise and intermittency of IMFs in the EMD process, and

this result has proved by substantial previous studies ([Yu et al., 2016](#); [Yu et al., 2015](#); [Ding, 2018](#)). It is interesting to note that the emerging MEEMD applied in this paper is superior to EEMD. This is mainly because MEEMD can remove some mode splitting caused by EEMD when applied in crude oil price decomposition ([Lang et al., 2020](#)). In evaluating the performance of the proposed hybrid model across multiple forecasting horizons, it is evident that its superiority remains robust for all horizons. However, the overall accuracy tends to decline as the forecasting horizon extends. For instance, the RMSE of our proposed model is minimized across all models and horizons, with values of 1.5717 for one-step-ahead forecasting, 2.4358 for two-step-ahead, and 2.6732 for three-step-ahead forecasts. This decline in overall accuracy is a common phenomenon observed in forecast models ([Jiang et al., 2022](#); [Fye et al., 2013](#)), stemming from the increased forecasting complexity and uncertainty in the longer terms forecasting.

To further corroborate the effectiveness of the proposed hybrid model, i.e., MEEMD-Mix-KELM, both DM test and MCS test are conducted on each comparative model group. In detail, the tests are performed in three groups, which is consistent with the forecasting experiment above. In terms of the DM test, the associated computational outcomes are presented in [Tables 7-9](#) (for horizon = 1, 2, and 3, respectively), where the first column presents the tested model, and the subsequent columns report the DM values along with their corresponding *p*-values (in brackets). As for the MCS test, the test results are presented in [Table 10](#), showcasing the calculated *p* values for different models using the TMAX and TR methods across horizons 1, 2, and 3. It is essential to emphasize that both the “*p*-value” in the DM test and the “*p* value” in the MCS test are constrained within the range of 0 to 1. In the DM test, a smaller “*p*-value” signifies an increased tendency to reject the null hypothesis, while in the MCS test, a larger “*p* value” is indicative of superior predictive performance of the model. From a statistical perspective, the conclusions drawn above are further supported by the results of the DM test and the MCS test. First, focusing on the single forecasting model (shown in panel A), all DM tests associated with KELM are consistently rejected at a 90% confidence level across various horizons, and all *p* values of KELM remains at 1 in the MCS test, supporting that KELM performs best among all single models. Second, both of the

Table 7

DM test for one-step-ahead predictions across various models.

Tested models	Benchmark models			
	SVR	ARIMA	BPNN	RW
Panel A: DM test for single forecasting models				
KELM	1.8951 (0.0290)	1.6083 (0.0539)	1.7043 (0.0442)	2.2398 (0.0126)
SVR		1.7254 (0.0422)	1.3118 (0.0948)	2.5028 (0.0062)
ARIMA			0.0755 (0.4699)	1.9689 (0.0245)
BPNN				2.0798 (0.0188)
Panel B: DM test for KELM models				
	RBF-KELM			SIG-KELM
Mix-KELM	1.8122 (0.0350)			2.2799 (0.0113)
Panel C: DM test for decomposition-ensemble models				
	EEMD-Mix-KELM	EMD-Mix-KELM	Mix-KELM	
MEEEMD-Mix-KELM	1.9245 (0.0271)	4.7898 (0.0000)	4.1818 (0.0000)	
EEMD-Mix-KELM		3.4359 (0.0003)	3.8665 (0.0001)	
EMD-Mix-KELM			3.4157 (0.0003)	

Table 8

DM test for two-step-ahead predictions across various models.

Tested models	Benchmark models			
	SVR	ARIMA	BPNN	RW
Panel A: DM test for single forecasting models				
KELM	3.2366 (0.0006)	2.6335 (0.0042)	2.5158 (0.0059)	1.9986 (0.0228)
SVR		1.5993 (0.0549)	2.1133 (0.0173)	1.7391 (0.0410)
ARIMA			−0.0505 (0.4799)	1.3144 (0.0944)
BPNN				1.2050 (0.1141)
Panel B: DM test for KELM models				
	RBF-KELM		SIG-KELM	
Mix-KELM	1.9261 (0.0270)		1.5292 (0.0631)	
Panel C: DM test for decomposition-ensemble models				
	EEMD-Mix-KELM	EMD-Mix-KELM	Mix-KELM	
MEEEMD-Mix-KELM	2.1619 (0.0153)	1.8163 (0.0347)	3.1068 (0.0009)	
EEMD-Mix-KELM		1.8341 (0.0333)	4.1430 (0.0000)	
EMD-Mix-KELM			3.0636 (0.0011)	

DM test and MCS test results (illustrated in panel B) confirm that our proposed Mix-KELM model is superior to the benchmark models with single global and local kernel. Third, in terms of tests for decomposition-ensemble models, we receive two useful results: on the one hand, decomposition-ensemble models statistically defeat the single forecasting model of KELM; on the other hand, the emerging MEEMD is statistically superior to the benchmark EMD family models.

Overall, the results demonstrate that our proposed MEEMD-Mix-KELM model, which incorporates the emerging MEEMD and improved Mix-KELM, outperforms other models in predicting WTI crude oil price across all horizons (ranging from 1 to 3), based on the MAE, SMAPE, and RMSE criteria. Moreover, the DM test and the MCS test provide statistical evidence to support the superiority of our proposed MEEMD-Mix-KELM model over previous EMD family-based models (EEMD-Mix-KELM and EMD-Mix-KELM) and single models (SVR, ARIMA, BPNN, SIG-ELM, and RBF-ELM). Taken together, the aforementioned findings provide strong support for the efficacy of our newly proposed MEEMD-Mix-KELM model as a highly promising tool for analyzing and forecasting crude oil prices.

Table 9

DM test for three-step-ahead predictions across various models.

Tested models	Benchmark models			
Panel A: DM test for single forecasting models				
	SVR	ARIMA	BPNN	RW
KELM	3.0226 (0.0013)	1.7079 (0.0438)	2.3912 (0.0084)	2.4643 (0.0069)
SVR		1.9417 (0.0261)	−2.0827 (0.0186)	2.2337 (0.0128)
ARIMA			0.0003 (0.4999)	2.2812 (0.0113)
BPNN				1.6433 (0.0502)
Panel B: DM test for KELM models				
	RBF-KELM		SIG-KELM	
Mix-KELM	1.6873 (0.0458)		1.7957 (0.0363)	
Panel C: DM test for decomposition-ensemble models				
	EEMD-Mix-KELM	EMD-Mix-KELM	Mix-KELM	
MEEMD-Mix-KELM	4.2427 (0.0000)	2.7056 (0.0034)	3.3257 (0.0004)	
EEMD-Mix-KELM		1.7872 (0.0370)	5.4161 (0.0000)	
EMD-Mix-KELM			3.1913 (0.0007)	

Table 10

The MCS test results.

MCS methods Forecast horizons	TMAX (p values)			TR (p values)		
	1-step-ahead	2-step-ahead	3-step-ahead	1-step-ahead	2-step-ahead	3-step-ahead
Panel A: MCS test for single forecasting models						
RW	0.017	0.055	0.070	0.051	0.022	0.080
BPNN	0.408	0.545	0.332	0.461	0.529	0.080
ARIMA	0.447	0.545	0.332	0.486	0.548	0.272
SVR	0.525	0.545	0.394	0.515	0.548	0.364
KELM	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>
Panel B: MCS test for KELM models						
SIG-KELM	0.732	0.404	0.454	0.177	0.248	0.262
RBF-KELM	0.732	0.404	0.454	0.214	0.337	0.262
Mix-KELM	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>
Panel C: MCS test for decomposition-ensemble models						
Mix-KELM	0.000	0.000	0.000	0.001	0.003	0.000
EMD-Mix-KELM	0.001	0.088	0.003	0.002	0.072	0.008
EEMD-Mix-KELM	0.104	0.114	0.056	0.094	0.096	0.077
MEEMD-Mix-KELM	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>	<u>1.000</u>

7.4. Further discussions

The proposed mix-kernel has been fully demonstrated to be a powerful tool to improve ELM forecasting by leveraging the strengths of the global kernel and local kernel. It is widely known that not only ELM, but the generalization ability of SVR is also strongly influenced by the selection of kernel function, which leads to the question of whether the mix-kernel devised for ELM in this study can be equally applicable to SVR. To address this, the mix-kernel, comprised of a linear combination of RBF and Sigmoid functions, is utilized as the kernel function for SVR. In particular, the parameters of RBF and Sigmoid kernels and the weights of the two kernels in the linear combination, are all determined by GA. Similarly, the Mix-SVR is compared with two benchmark SVR modes with single kernels, i.e., SIG-SVR and RBF-SVR. The comparison results are reported in Figs. 6–8.

A promising result is found that the prediction accuracy of single kernel-based SVRs is improved successfully with the help of our proposed mix-kernel. Specifically, in the case of SVR, the mix-kernel can

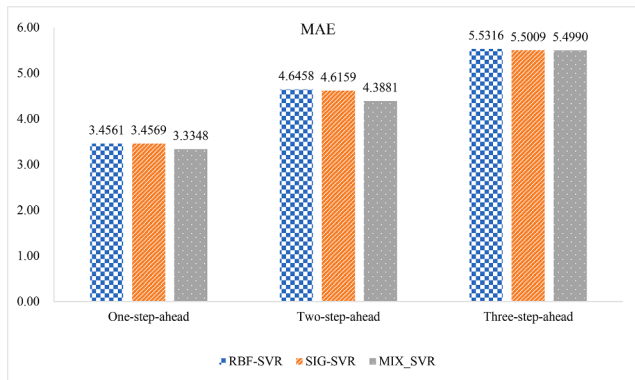


Fig. 6. MAE of various kernel-based SVRs.

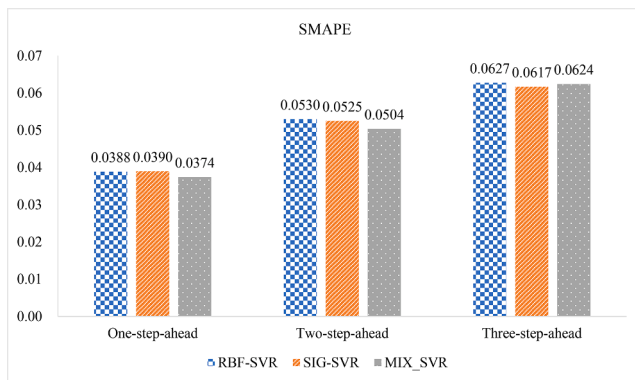


Fig. 7. SMAPE of various kernel-based SVRs.

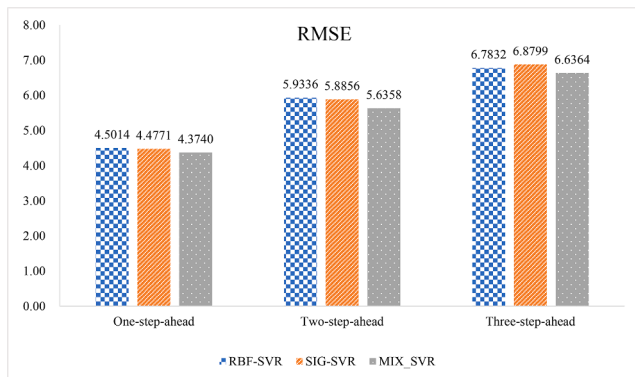


Fig. 8. RMSE of various kernel-based SVRs.

defeat the single RBF and Sigmoid in terms of all error criteria (i.e., MAE, SMAPE, RMSE) at all horizons. These rosy results further proved the powerful strength of our proposed mix-kernel, which has absorbed the advantages of the global kernel (in generation) and local kernel (in learning).

In addition to forecasting accuracy, computational efficiency plays a crucial role in developing a novel forecasting model, such as the MEEMD-Mix-KELM proposed in this paper. The computational times of the proposed model and its benchmarks are calculated and displayed in

Table 11. When comparing single models, there is no doubt that the naïve RW stands out as the most time-saving model, avoiding the complexities of parameter estimation and training processes inherent in other econometric and AI models. Additionally, the KELM model outperforms the other three single models (ARIMA, BPNN, SVR) in computational efficiency, aligning with findings by Biso et al. (2019). The reason is that the KELM randomly set the input weights and biases and calculate output weights analytically, leading to faster training times and makes it particularly efficient for large-scale datasets (Li and Zhao, 2020).

However, our investigation reveals that both the Mix-KELM and decomposition-based Mix-KELM models exhibit higher time consumption compared to the single KELM model. As for the Mix-KELM, the additional time is primarily attributed to the increased computational workload associated with optimizing weights through the use of GA. As for decomposition-based models, the extra time is incurred due to the necessity of decomposing the oil price and conducting separate predictions for each decomposed mode. Specifically, our proposed MEEMD-Mix-KELM demonstrates equally time consumption to EEMD-Mix-KELM but with superior accuracy performance. Both of the two hybrid models surpass the EMD-based hybrid model (i.e., EMD-Mix-KELM) in time consumption due to their ensemble nature, involving the repetition of the decomposition process multiple times with added white noise to enhance decomposition quality. Although the total computational time increases for our proposed hybrid model, it is still shorter than 3 min and would be acceptable for practical applications.

8. Conclusions

Given the complexity of the dynamics underlying global crude oil prices and the inherent uncertainty inherent in forecasting such prices, this paper puts forth a novel hybrid model, i.e., MEEMD-Mix-KELM, for the purpose of forecasting crude oil prices. The proposed methodology comprises three primary stages, namely oil price decomposition, mix-kernel design, and oil price forecasting. Firstly, an emerging decomposition method, i.e., MEEMD, which successfully solves the mode splitting problem of EEMD, is utilized to decompose the crude oil price into a set of simple subseries. Secondly, a novel mix-kernel, which is constructed by absorbing the advantage of the global kernel and local kernel with the help of GA, is particularly designed for ELM. Thirdly, the proposed MEEMD-Mix-KELM is employed to forecast the WTI crude oil price. In comparison to the pre-existing models, the innovative methodology proposed in this work contributes to the existing literature by (1) being the first attempt to adopt the nascent decomposition method MEEMD to crude oil price modeling and analysis and (2) constructing a novel mix-kernel for ELM to promote crude oil price prediction. The main conclusions of this study include,

- The emerging MEEMD approach utilized in this paper is superior to other EMD family models (i.e., EMD and EEMD) in effectively modeling various subseries of the WTI crude oil price, thereby improving the accuracy of subsequent forecasts.
- The proposed mix-kernel for ELM is a promising kernel that has both excellent extrapolation ability (learned from Sigmoid kernel) and outstanding interpolation ability (learned from RBF kernel), and presents an apparent advantage over single kernels for crude oil price forecasting.
- The newly proposed hybrid method namely MEEMD-Mix-KELM exhibits exceptional forecasting capabilities for the crude oil price in

Table 11

The computational time of the proposed model and its benchmarks for one-step-ahead predictions.

Models	RW	ARIMA	BPNN	SVR	KELM	Mix-KELM	EMD-Mix-KELM	EEMD-Mix-KELM	MEEMD-Mix-KELM
Computational time (s)	1.67	23.23	13.05	12.04	10.28	23.80	103.68	170.82	173.31

terms of all error criteria and forecasting horizons, whose robustness and reliability have been fully verified by the DM test and MCS test.

- We also identified potential opportunities of using the proposed mix-kernel for SVR, which further confirms the efficacy of our proposed mix-kernel method.

Overall, the aforementioned results collectively underscore the exceptional forecasting prowess of the novel framework proposed in this study for the crude oil prices. The accurate forecasting results obtained from our proposed model offer significant benefits to various stakeholders who are interested in the crude oil market, including investors, regulators, and academics. Specifically, our model supports investors in portfolio optimization, aids regulators in assessing macroeconomic risks, and assists academics in advancing knowledge within energy economics and forecasting methodologies. Nevertheless, it is worth noting that there is still scope for further improvement. Firstly, beyond the WTI crude oil market, this highly-effective forecasting framework may be readily extendable to other international crude oil markets and closely-linked financial markets, such as the gold market. Secondly, while this study considered only two typical local and global kernels (RBF and Sigmoid) in constructing the mix-kernel, other powerful kernels (as listed in Table 1) could also be incorporated to improve the accuracy of predictions even further.

Author contributions

Jingjing Li and Chengyuan Zhang conceived of the presented idea. Jingjing Li and Zhanjiang Hong developed the integrated analysis framework and performed the experiment. Jingjing Li and Chengyuan Zhang collected and analyzed the data. Jingjing Li, Zhanjiang Hong, Jiaqian Wu and Cuicui Yu contributed to the interpretation of the results. Jingjing Li took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis and manuscript. All authors read and approved the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2023.123104>.

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